INTERVIEW PREPARATION

DSA Coding Prep Solutions Guide

A complete set of solved coding questions in JavaScript, organized by topic — from programming basics through arrays. Written with simple logic and clear variable names, so every solution is easy to follow and easy to explain in an interview.

110 QUESTIONS

5 TOPICS

JS LANGUAGE

Basics of Programming

Q1 Sum of Two Integers

EASY

```
let a = 5;
let b = 10;
let sum = a + b;
console.log("Sum is:", sum);
```

Q2 Relation Between Integer and String

EASY

Shows how JS converts a number when it is joined with a string (type coercion).

```
let num = 5;
let text = "5";

console.log(num + 1);        // 6   (number + number)
console.log(text + 1);       // "51" (string + number becomes string)
console.log(num == text);    // true  (loose equality checks value only)
console.log(num === text);   // false (strict equality checks type too)
```

Q3 Sum and Message

EASY

```
let a = 12;
let b = 8;
let sum = a + b;
console.log("The sum of " + a + " and " + b + " is " + sum);
```

Q4 Accept and Print the Answer

EASY

Using readline in Node.js to take input from the user.

```
const readline = require("readline-sync");

let name = readline.question("Enter your name: ");
let age = readline.question("Enter your age: ");

console.log("Hello " + name + ", you are " + age + " years old.");
```

Q5 Swap Two Variables (3 Methods)

EASY

```
// Method 1: using a third variable
let a = 5, b = 10, temp;
temp = a;
a = b;
b = temp;
console.log(a, b);

// Method 2: using arithmetic (no third variable)
let x = 5, y = 10;
x = x + y;
y = x - y;
x = x - y;
console.log(x, y);

// Method 3: using array destructuring (simplest way in JS)
let p = 5, q = 10;
[p, q] = [q, p];
console.log(p, q);
```

Q6 Calculate Compound Interest

EASY

```
let principal = 10000;
let rate = 5;      // percent per year
let time = 3;       // years

let amount = principal * Math.pow((1 + rate / 100), time);
let compoundInterest = amount - principal;

console.log("Compound Interest:", compoundInterest.toFixed(2));
```

Q7 Generate OTP

EASY

```
function generateOTP(length) {
  let otp = "";
  for (let i = 0; i < length; i++) {
    let digit = Math.floor(Math.random() * 10); // random number 0-9
    otp += digit;
  }
  return otp;
}

console.log("Your OTP is:", generateOTP(6));
```

Q8 Circumference of Circle

EASY

```
let radius = 7;
let circumference = 2 * Math.PI * radius;
console.log("Circumference:", circumference.toFixed(2));
```

Q9 Swap two numbers without using a third variable

IMPORTANT

```
let a = 15, b = 25;
a = a + b;
b = a - b;
a = a - b;
console.log("a =", a, "b =", b);
```

Q10 Celsius to Fahrenheit conversion

IMPORTANT

```
let celsius = 37;
let fahrenheit = (celsius * 9 / 5) + 32;
console.log(celsius + "°C is " + fahrenheit + "°F");
```

Q11 Fahrenheit to Celsius conversion

IMPORTANT

```
let fahrenheit = 98.6;
let celsius = (fahrenheit - 32) * 5 / 9;
console.log(fahrenheit + "°F is " + celsius.toFixed(2) + "°C");
```

Q12 Area and Perimeter of Rectangle

IMPORTANT

```
let length = 8;
let width = 5;

let area = length * width;
let perimeter = 2 * (length + width);

console.log("Area:", area);
console.log("Perimeter:", perimeter);
```

Q13 Simple Interest

IMPORTANT

```
let principal = 5000;
let rate = 6;
let time = 2;

let simpleInterest = (principal * rate * time) / 100;
console.log("Simple Interest:", simpleInterest);
```

Q14 Calculate Percentage and Grade

IMPORTANT

```
let marks = [85, 90, 78, 92, 88]; // marks in 5 subjects
let total = marks.reduce((sum, m) => sum + m, 0);
let percentage = total / marks.length;

let grade;
if (percentage >= 90) grade = "A+";
else if (percentage >= 75) grade = "A";
else if (percentage >= 60) grade = "B";
else if (percentage >= 40) grade = "C";
else grade = "Fail";

console.log("Percentage:", percentage.toFixed(2));
console.log("Grade:", grade);
```

Q15 ASCII value of a character

IMPORTANT

```
let ch = "A";
let asciiValue = ch.charCodeAt(0);
console.log("ASCII value of " + ch + " is " + asciiValue);

// reverse: number to character
let code = 97;
console.log(String.fromCharCode(code)); // "a"
```

Q16 Convert days into years, months and days

IMPORTANT

```
let totalDays = 800;

let years = Math.floor(totalDays / 365);
let remainingAfterYears = totalDays % 365;

let months = Math.floor(remainingAfterYears / 30);
let days = remainingAfterYears % 30;

console.log(years + " years, " + months + " months, " + days + " days");
```

Q17 Area of Triangle using Heron's Formula

MEDIUM

```
let sideA = 5, sideB = 6, sideC = 7;

let s = (sideA + sideB + sideC) / 2; // semi-perimeter
let area = Math.sqrt(s * (s - sideA) * (s - sideB) * (s - sideC));

console.log("Area of triangle:", area.toFixed(2));
```

TOPIC 02

Control Flow (if-else)

Q18 Greatest Between Two Numbers

EASY

```
let a = 15, b = 22;

if (a > b) {
  console.log(a, "is greater");
} else {
  console.log(b, "is greater");
}
```

Q19 Check Even or Odd

EASY

```
let num = 17;

if (num % 2 === 0) {
  console.log(num, "is Even");
} else {
  console.log(num, "is Odd");
}
```

Q20 Greatest Among Three Numbers

EASY

```
let a = 12, b = 45, c = 30;

if (a >= b && a >= c) {
  console.log(a, "is the greatest");
} else if (b >= a && b >= c) {
  console.log(b, "is the greatest");
} else {
  console.log(c, "is the greatest");
}
```

Q21 Leap Year Check

EASY

```
let year = 2024;

if ((year % 4 === 0 && year % 100 !== 0) || year % 400 === 0) {
  console.log(year, "is a Leap Year");
} else {
  console.log(year, "is not a Leap Year");
}
```

Q22 Check Positive, Negative or Zero

IMPORTANT

```
let num = -8;

if (num > 0) {
  console.log("Positive");
} else if (num < 0) {
  console.log("Negative");
} else {
  console.log("Zero");
}
```

Q23 Check Divisibility by 5 and 11

IMPORTANT

```
let num = 55;

if (num % 5 === 0 && num % 11 === 0) {
  console.log(num, "is divisible by both 5 and 11");
} else {
  console.log(num, "is not divisible by both");
}
```

Q24 Find Absolute Value

IMPORTANT

```
let num = -25;
let absValue = num < 0 ? -num : num; // or use Math.abs(num)
console.log("Absolute value:", absValue);
```

Q25 Check Character is Alphabet/Digit/Special Character

IMPORTANT

```
let ch = "@";

if ((ch >= "a" && ch <= "z") || (ch >= "A" && ch <= "Z")) {
  console.log(ch, "is an Alphabet");
} else if (ch >= "0" && ch <= "9") {
  console.log(ch, "is a Digit");
} else {
  console.log(ch, "is a Special Character");
}
```

Q26 Check Uppercase or Lowercase

IMPORTANT

```
let ch = "k";

if (ch >= "A" && ch <= "Z") {
  console.log(ch, "is Uppercase");
} else if (ch >= "a" && ch <= "z") {
  console.log(ch, "is Lowercase");
} else {
  console.log(ch, "is not an alphabet");
}
```

Q27 Find Maximum of Four Numbers

IMPORTANT

```
let a = 10, b = 55, c = 33, d = 47;

let max = a;
if (b > max) max = b;
if (c > max) max = c;
if (d > max) max = d;

console.log("Maximum number is:", max);

// simple alternative
console.log(Math.max(a, b, c, d));
```

Q28 Voter Eligibility Check

MEDIUM

```
let age = 17;
let hasVoterId = true;

if (age >= 18 && hasVoterId) {
  console.log("Eligible to vote");
} else if (age < 18) {
  console.log("Not eligible, age is below 18");
} else {
  console.log("Not eligible, voter ID missing");
}
```

Q29 Shop Discount Calculation

MEDIUM

```
let billAmount = 2500;
let discountPercent;

if (billAmount >= 2000) {
  discountPercent = 20;
} else if (billAmount >= 1000) {
  discountPercent = 10;
} else {
  discountPercent = 0;
}

let discount = (billAmount * discountPercent) / 100;
let finalAmount = billAmount - discount;

console.log("Discount:", discount);
console.log("Final Amount to pay:", finalAmount);
```

Q30 Electricity Bill Calculation

MEDIUM

```
let units = 250;
let bill = 0;

if (units <= 100) {
  bill = units * 3;
} else if (units <= 200) {
  bill = 100 * 3 + (units - 100) * 4;
} else {
  bill = 100 * 3 + 100 * 4 + (units - 200) * 6;
}

console.log("Electricity Bill: Rs.", bill);
```

Q31 Count Number of Days in a Month

MEDIUM

```
let month = 2;
let year = 2024;
let days;

if ([1, 3, 5, 7, 8, 10, 12].includes(month)) {
  days = 31;
} else if ([4, 6, 9, 11].includes(month)) {
  days = 30;
} else {
  // February - check leap year
  days = ((year % 4 === 0 && year % 100 !== 0) || year % 400 === 0) ? 29 : 28;
}

console.log("Number of days:", days);
```

TOPIC 03

Loops

Q32 Print "Hello World" N Times

EASY

```
let n = 5;
for (let i = 1; i <= n; i++) {
  console.log("Hello World");
}
```

Q33 Print Natural Numbers up to N

EASY

```
let n = 10;
for (let i = 1; i <= n; i++) {
  console.log(i);
}
```

Q34 Print Numbers from N to 1

EASY

```
let n = 10;
for (let i = n; i >= 1; i--) {
  console.log(i);
}
```

Q35 Multiplication Table

EASY

```
let num = 7;
for (let i = 1; i <= 10; i++) {
  console.log(num + " x " + i + " = " + (num * i));
}
```

Q36 Sum of First N Numbers

EASY

```
let n = 10;
let sum = 0;
for (let i = 1; i <= n; i++) {
  sum += i;
}
console.log("Sum:", sum);
```

Q37 Factorial

EASY

```
let n = 5;
let factorial = 1;
for (let i = 1; i <= n; i++) {
  factorial *= i;
}
console.log("Factorial of " + n + " is " + factorial);
```

Q38 Sum of Even and Odd Numbers Separately

EASY

```
let n = 10;
let evenSum = 0, oddSum = 0;

for (let i = 1; i <= n; i++) {
  if (i % 2 === 0) evenSum += i;
  else oddSum += i;
}

console.log("Even Sum:", evenSum);
console.log("Odd Sum:", oddSum);
```

Q39 Check Vowel/Consonant using Switch

EASY

```
let ch = "e";

switch (ch.toLowerCase()) {
  case "a":
  case "e":
  case "i":
  case "o":
  case "u":
    console.log(ch, "is a Vowel");
    break;
  default:
    console.log(ch, "is a Consonant");
}
```

Q40 Repeat Hello (do-while)

EASY

```
let n = 5;
let i = 1;

do {
  console.log("Hello");
  i++;
} while (i <= n);
```

Q41 Fibonacci Series

IMPORTANT

```
let n = 10;
let first = 0, second = 1;

for (let i = 1; i <= n; i++) {
  console.log(first);
  let next = first + second;
  first = second;
  second = next;
}
```

Q42 Armstrong Number

IMPORTANT

A number equal to the sum of its own digits, each raised to the power of the digit count.

```
let num = 153;
let temp = num;
let digitCount = String(num).length;
let sum = 0;

while (temp > 0) {
  let digit = temp % 10;
  sum += Math.pow(digit, digitCount);
  temp = Math.floor(temp / 10);
}

if (sum === num) console.log(num, "is an Armstrong Number");
else console.log(num, "is not an Armstrong Number");
```

Q43 Perfect Number

IMPORTANT

A number equal to the sum of its divisors (excluding itself). Example: 28 = 1+2+4+7+14.

```
let num = 28;
let sum = 0;

for (let i = 1; i < num; i++) {
  if (num % i === 0) sum += i;
}

if (sum === num) console.log(num, "is a Perfect Number");
else console.log(num, "is not a Perfect Number");
```

Q44 Strong Number

IMPORTANT

A number equal to the sum of the factorial of its digits. Example: 145 = 1! + 4! + 5!.

```
function factorial(n) {
  let f = 1;
  for (let i = 1; i <= n; i++) f *= i;
  return f;
}

let num = 145;
let temp = num;
let sum = 0;

while (temp > 0) {
  let digit = temp % 10;
  sum += factorial(digit);
  temp = Math.floor(temp / 10);
}

if (sum === num) console.log(num, "is a Strong Number");
else console.log(num, "is not a Strong Number");
```

Q45 Neon Number

IMPORTANT

A number whose square's digits add up to the number itself. Example: 9, square = 81, 8+1 = 9.

```
let num = 9;
let square = num * num;
let sum = 0;

while (square > 0) {
  sum += square % 10;
  square = Math.floor(square / 10);
}

if (sum === num) console.log(num, "is a Neon Number");
else console.log(num, "is not a Neon Number");
```

Q46 Harshad Number

IMPORTANT

A number divisible by the sum of its own digits.

```
let num = 18;
let temp = num;
let digitSum = 0;

while (temp > 0) {
  digitSum += temp % 10;
  temp = Math.floor(temp / 10);
}

if (num % digitSum === 0) console.log(num, "is a Harshad Number");
else console.log(num, "is not a Harshad Number");
```

Q47 Count Digits

IMPORTANT

```
let num = 45892;
let temp = num;
let count = 0;

while (temp > 0) {
  count++;
  temp = Math.floor(temp / 10);
}

console.log("Number of digits:", count);
```

Q48 Product of Digits

IMPORTANT

```
let num = 1234;
let temp = num;
let product = 1;

while (temp > 0) {
  let digit = temp % 10;
  product *= digit;
  temp = Math.floor(temp / 10);
}

console.log("Product of digits:", product);
```

Q49 Reverse Digits

IMPORTANT

```
let num = 12345;
let temp = num;
let reversed = 0;

while (temp > 0) {
  let digit = temp % 10;
  reversed = reversed * 10 + digit;
  temp = Math.floor(temp / 10);
}

console.log("Reversed number:", reversed);
```

Q50 Sum of Squares of Digits

IMPORTANT

```
let num = 123;
let temp = num;
let sum = 0;

while (temp > 0) {
  let digit = temp % 10;
  sum += digit * digit;
  temp = Math.floor(temp / 10);
}

console.log("Sum of squares of digits:", sum);
```

Q51 Print Prime Numbers in a Range

IMPORTANT

```
function isPrime(num) {
  if (num < 2) return false;
  for (let i = 2; i <= Math.sqrt(num); i++) {
    if (num % i === 0) return false;
  }
  return true;
}

let start = 10, end = 50;
for (let num = start; num <= end; num++) {
  if (isPrime(num)) console.log(num);
}
```

Q52 LCM of Two Numbers

IMPORTANT

```
let a = 12, b = 18;
let greater = a > b ? a : b;

while (true) {
  if (greater % a === 0 && greater % b === 0) {
    console.log("LCM is:", greater);
    break;
  }
  greater++;
}
```

Q53 GCD of Two Numbers

IMPORTANT

```
let a = 36, b = 24;

while (b !== 0) {
  let temp = b;
  b = a % b;
  a = temp;
}

console.log("GCD is:", a);
```

Q54 Print All Factors

MEDIUM

```
let num = 36;

for (let i = 1; i <= num; i++) {
  if (num % i === 0) console.log(i);
}
```

Q55 Prime Number Check

MEDIUM

```
let num = 29;
let isPrime = true;

if (num < 2) {
  isPrime = false;
} else {
  for (let i = 2; i <= Math.sqrt(num); i++) {
    if (num % i === 0) {
      isPrime = false;
      break;
    }
  }
}

console.log(num, isPrime ? "is Prime" : "is not Prime");
```

Q56 Power (a^b)

MEDIUM

```
let base = 2, exponent = 10;
let result = 1;

for (let i = 1; i <= exponent; i++) {
  result *= base;
}

console.log(base + "^" + exponent + " = " + result);
// simple alternative: Math.pow(base, exponent)
```

Q57 Sum of Digits

MEDIUM

```
let num = 9875;
let temp = num;
let sum = 0;

while (temp > 0) {
  sum += temp % 10;
  temp = Math.floor(temp / 10);
}

console.log("Sum of digits:", sum);
```

Q58 Reverse a Number

MEDIUM

```
let num = -5321;
let sign = num < 0 ? -1 : 1;
let temp = Math.abs(num);
let reversed = 0;

while (temp > 0) {
  reversed = reversed * 10 + (temp % 10);
  temp = Math.floor(temp / 10);
}

console.log("Reversed:", reversed * sign);
```

Q59 ISBN Number Check

MEDIUM

A 10-digit ISBN is valid if the weighted sum (digit * position) is divisible by 11.

```
let isbn = "0306406152";
let sum = 0;

for (let i = 0; i < 10; i++) {
  let digit = Number(isbn[i]);
  sum += digit * (i + 1);
}

if (sum % 11 === 0) console.log("Valid ISBN");
else console.log("Invalid ISBN");
```

Q60 Automorphic Number

MEDIUM

A number whose square ends with the number itself. Example: 25, square = 625, ends with 25.

```
let num = 25;
let square = num * num;

let numStr = String(num);
let squareStr = String(square);

if (squareStr.endsWith(numStr)) console.log(num, "is Automorphic");
else console.log(num, "is not Automorphic");
```

Q61 Area Calculator using Switch

MEDIUM

```
let shape = "circle";
let area;

switch (shape) {
  case "circle":
    let radius = 7;
    area = Math.PI * radius * radius;
    break;
  case "square":
    let side = 5;
    area = side * side;
    break;
  case "rectangle":
    let length = 6, width = 4;
    area = length * width;
    break;
  default:
    area = 0;
}

console.log("Area:", area.toFixed(2));
```

Q62 Guess the Number

MEDIUM

```
let secretNumber = 42;
let guesses = [50, 30, 42]; // simulated user guesses
let attempt = 0;
let found = false;

while (attempt < guesses.length && !found) {
  let guess = guesses[attempt];
  attempt++;

  if (guess === secretNumber) {
    console.log("Correct! You guessed it in " + attempt + " tries");
    found = true;
  } else if (guess < secretNumber) {
    console.log("Too low, try again");
  } else {
    console.log("Too high, try again");
  }
}
```

Q63 Sasta Calculator

MEDIUM

```
function calculator(a, b, operator) {
  switch (operator) {
    case "+": return a + b;
    case "-": return a - b;
    case "*": return a * b;
    case "/": return b !== 0 ? a / b : "Cannot divide by zero";
    default: return "Invalid operator";
  }
}

console.log(calculator(10, 5, "+"));
console.log(calculator(10, 5, "*"));
```

Q64 Restaurant Menu Project

MEDIUM

```
let menu = {
  "burger": 120,
  "pizza": 250,
  "pasta": 180,
  "coldDrink": 50
};

let order = ["burger", "pizza", "coldDrink"];
let total = 0;

for (let i = 0; i < order.length; i++) {
  let item = order[i];
  let price = menu[item];
  console.log(item + " - Rs." + price);
  total += price;
}

console.log("Total bill: Rs.", total);
```

TOPIC 04

Pattern Printing

Q65 Right Angle Star Pattern

EASY

n = 5

```
*
* *
* * *
* * * *
* * * * *
```

```
let n = 5;
for (let i = 1; i <= n; i++) {
  let row = "";
  for (let j = 1; j <= i; j++) {
    row += "* ";
  }
  console.log(row);
}
```

Q66 Right Angle Number Pattern

EASY

```
let n = 5;
for (let i = 1; i <= n; i++) {
  let row = "";
  for (let j = 1; j <= i; j++) {
    row += j + " ";
  }
  console.log(row);
}
```

Q67 Right Angle Alphabet Pattern

EASY

```
let n = 5;
for (let i = 1; i <= n; i++) {
  let row = "";
  for (let j = 0; j < i; j++) {
    row += String.fromCharCode(65 + j) + " "; // 65 = 'A'
  }
  console.log(row);
}
```

Q68 Square Pattern

IMPORTANT

```
let n = 5;
for (let i = 1; i <= n; i++) {
  let row = "";
  for (let j = 1; j <= n; j++) {
    row += "* ";
  }
  console.log(row);
}
```

Q69 Hollow Square

IMPORTANT

```
let n = 5;
for (let i = 1; i <= n; i++) {
  let row = "";
  for (let j = 1; j <= n; j++) {
    if (i === 1 || i === n || j === 1 || j === n) {
      row += "* ";
    } else {
      row += "  ";
    }
  }
  console.log(row);
}
```

Q70 Hollow Rectangle

IMPORTANT

```
let rows = 5, cols = 9;
for (let i = 1; i <= rows; i++) {
  let row = "";
  for (let j = 1; j <= cols; j++) {
    if (i === 1 || i === rows || j === 1 || j === cols) {
      row += "*";
    } else {
      row += " ";
    }
  }
  console.log(row);
}
```

Q71 Floyd's Triangle

IMPORTANT

```
let n = 5;
let num = 1;

for (let i = 1; i <= n; i++) {
  let row = "";
  for (let j = 1; j <= i; j++) {
    row += num + " ";
    num++;
  }
  console.log(row);
}
```

Q72 Pascal Triangle

IMPORTANT

```
let n = 5;

for (let i = 0; i < n; i++) {
  let row = "";
  let value = 1;
  for (let j = 0; j <= i; j++) {
    row += value + " ";
    value = (value * (i - j)) / (j + 1);
  }
  console.log(row);
}
```

Q73 Diamond Pattern

IMPORTANT

```
let n = 4;

// upper half (including middle)
for (let i = 1; i <= n; i++) {
  let spaces = " ".repeat(n - i);
  let stars = "*".repeat(2 * i - 1);
  console.log(spaces + stars);
}

// lower half
for (let i = n - 1; i >= 1; i--) {
  let spaces = " ".repeat(n - i);
  let stars = "*".repeat(2 * i - 1);
  console.log(spaces + stars);
}
```

Q74 Hollow Diamond

IMPORTANT

```
let n = 4;

// upper half
for (let i = 1; i <= n; i++) {
  let spaces = " ".repeat(n - i);
  if (i === 1) {
    console.log(spaces + "*");
  } else {
    console.log(spaces + "*" + " ".repeat(2 * i - 3) + "*");
  }
}

// lower half
for (let i = n - 1; i >= 1; i--) {
  let spaces = " ".repeat(n - i);
  if (i === 1) {
    console.log(spaces + "*");
  } else {
    console.log(spaces + "*" + " ".repeat(2 * i - 3) + "*");
  }
}
```

Q75 Butterfly Pattern

IMPORTANT

```
let n = 4;

// upper half
for (let i = 1; i <= n; i++) {
  let stars = "*".repeat(i);
  let spaces = " ".repeat(2 * (n - i));
  console.log(stars + spaces + stars);
}

// lower half
for (let i = n; i >= 1; i--) {
  let stars = "*".repeat(i);
  let spaces = " ".repeat(2 * (n - i));
  console.log(stars + spaces + stars);
}
```

Q76 Pyramid of Numbers

IMPORTANT

```
let n = 5;
for (let i = 1; i <= n; i++) {
  let spaces = " ".repeat(n - i);
  let numbers = "";
  for (let j = 1; j <= i; j++) numbers += j + " ";
  console.log(spaces + numbers);
}
```

Q77 Pyramid of Alphabets

IMPORTANT

```
let n = 5;
for (let i = 1; i <= n; i++) {
  let spaces = " ".repeat(n - i);
  let letters = "";
  for (let j = 0; j < i; j++) {
    letters += String.fromCharCode(65 + j) + " ";
  }
  console.log(spaces + letters);
}
```

Q78 Inverted Right Angle Triangle

MEDIUM

```
let n = 5;
for (let i = n; i >= 1; i--) {
  let row = "";
  for (let j = 1; j <= i; j++) {
    row += "* ";
  }
  console.log(row);
}
```

Q79 Mirror Right Angle Triangle

MEDIUM

```
let n = 5;
for (let i = 1; i <= n; i++) {
  let spaces = " ".repeat(n - i);
  let stars = "*".repeat(i);
  console.log(spaces + stars);
}
```

Q80 Centered Triangle Pattern

MEDIUM

```
let n = 5;
for (let i = 1; i <= n; i++) {
  let spaces = " ".repeat(n - i);
  let stars = "*".repeat(2 * i - 1);
  console.log(spaces + stars);
}
```

Q81 V Pattern

MEDIUM

```
let n = 5;
for (let i = 0; i < n; i++) {
  let leftSpaces = " ".repeat(i);
  let rightSpaces = " ".repeat(2 * (n - i - 1));
  console.log(leftSpaces + "*" + rightSpaces + "*");
}
```

Q82 X Pattern

MEDIUM

```
let n = 5;
for (let i = 0; i < n; i++) {
  let row = "";
  for (let j = 0; j < n; j++) {
    if (j === i || j === n - i - 1) {
      row += "*";
    } else {
      row += " ";
    }
  }
  console.log(row);
}
```

TOPIC 05

Arrays (Level-1)

Q83 Sum and Average of Array Elements

EASY

```
let arr = [10, 20, 30, 40, 50];
let sum = 0;

for (let i = 0; i < arr.length; i++) {
  sum += arr[i];
}

let average = sum / arr.length;

console.log("Sum:", sum);
console.log("Average:", average);
```

Q84 Find Greatest Element

EASY

```
let arr = [12, 45, 3, 67, 22];
let greatest = arr[0];

for (let i = 1; i < arr.length; i++) {
  if (arr[i] > greatest) {
    greatest = arr[i];
  }
}

console.log("Greatest element:", greatest);
```

Q85 Reverse an Array

EASY

```
let arr = [1, 2, 3, 4, 5];
let start = 0;
let end = arr.length - 1;

while (start < end) {
  let temp = arr[start];
  arr[start] = arr[end];
  arr[end] = temp;
  start++;
  end--;
}

console.log("Reversed array:", arr);
```

Q86 Left Rotation by One

EASY

```
let arr = [1, 2, 3, 4, 5];
let first = arr[0];

for (let i = 0; i < arr.length - 1; i++) {
  arr[i] = arr[i + 1];
}
arr[arr.length - 1] = first;

console.log("After left rotation by 1:", arr);
```

Q87 Linear Search

EASY

```
let arr = [4, 8, 15, 16, 23, 42];
let target = 16;
let foundIndex = -1;

for (let i = 0; i < arr.length; i++) {
  if (arr[i] === target) {
    foundIndex = i;
    break;
  }
}

if (foundIndex !== -1) console.log("Found at index:", foundIndex);
else console.log("Not found");
```

Q88 Binary Search

EASY

Array must be sorted before applying binary search.

```
let arr = [2, 5, 8, 12, 16, 23, 38, 45];
let target = 23;

let low = 0;
let high = arr.length - 1;
let foundIndex = -1;

while (low <= high) {
  let mid = Math.floor((low + high) / 2);

  if (arr[mid] === target) {
    foundIndex = mid;
    break;
  } else if (arr[mid] < target) {
    low = mid + 1;
  } else {
    high = mid - 1;
  }
}

console.log(foundIndex !== -1 ? "Found at index " + foundIndex : "Not found");
```

Q89 Find Minimum Element

IMPORTANT

```
let arr = [12, 45, 3, 67, 22];
let smallest = arr[0];

for (let i = 1; i < arr.length; i++) {
  if (arr[i] < smallest) {
    smallest = arr[i];
  }
}

console.log("Smallest element:", smallest);
```

Q90 Count Even and Odd Elements

IMPORTANT

```
let arr = [1, 2, 3, 4, 5, 6, 7, 8];
let evenCount = 0, oddCount = 0;

for (let i = 0; i < arr.length; i++) {
  if (arr[i] % 2 === 0) evenCount++;
  else oddCount++;
}

console.log("Even count:", evenCount);
console.log("Odd count:", oddCount);
```

Q91 Copy One Array to Another

IMPORTANT

```
let source = [1, 2, 3, 4, 5];
let destination = [];

for (let i = 0; i < source.length; i++) {
  destination[i] = source[i];
}

console.log("Copied array:", destination);
```

Q92 Merge Two Arrays

IMPORTANT

```
let arr1 = [1, 2, 3];
let arr2 = [4, 5, 6];
let merged = [];

for (let i = 0; i < arr1.length; i++) merged.push(arr1[i]);
for (let i = 0; i < arr2.length; i++) merged.push(arr2[i]);

console.log("Merged array:", merged);
```

Q93 Check if Array is Sorted

IMPORTANT

```
let arr = [1, 2, 3, 4, 5];
let isSorted = true;

for (let i = 0; i < arr.length - 1; i++) {
  if (arr[i] > arr[i + 1]) {
    isSorted = false;
    break;
  }
}

console.log(isSorted ? "Array is sorted" : "Array is not sorted");
```

Q94 Insert Element at Given Position

IMPORTANT

```
let arr = [1, 2, 4, 5];
let position = 2;   // index where we want to insert
let value = 3;

arr.splice(position, 0, value);

console.log("After insertion:", arr);
```

Q95 Delete Element from Array

IMPORTANT

```
let arr = [10, 20, 30, 40, 50];
let position = 2; // index of element to delete

arr.splice(position, 1);

console.log("After deletion:", arr);
```

Q96 Find Missing Number in Array

IMPORTANT

Array contains numbers from 1 to n with one number missing.

```
let arr = [1, 2, 4, 5, 6];
let n = arr.length + 1; // since one number is missing

let expectedSum = (n * (n + 1)) / 2;
let actualSum = 0;

for (let i = 0; i < arr.length; i++) {
  actualSum += arr[i];
}

let missingNumber = expectedSum - actualSum;
console.log("Missing number:", missingNumber);
```

Q97 Find Duplicate Elements

IMPORTANT

```
let arr = [1, 2, 3, 2, 4, 5, 1];
let seen = {};
let duplicates = [];

for (let i = 0; i < arr.length; i++) {
  let value = arr[i];
  if (seen[value]) {
    duplicates.push(value);
  } else {
    seen[value] = true;
  }
}

console.log("Duplicate elements:", duplicates);
```

Q98 Remove Duplicate Elements (Unsorted)

IMPORTANT

```
let arr = [1, 2, 3, 2, 4, 5, 1];
let seen = {};
let result = [];

for (let i = 0; i < arr.length; i++) {
  let value = arr[i];
  if (!seen[value]) {
    seen[value] = true;
    result.push(value);
  }
}

console.log("Array without duplicates:", result);
```

Q99 Find Second Greatest Element

MEDIUM

```
let arr = [12, 45, 3, 67, 22, 67];
let greatest = -Infinity;
let secondGreatest = -Infinity;

for (let i = 0; i < arr.length; i++) {
  if (arr[i] > greatest) {
    secondGreatest = greatest;
    greatest = arr[i];
  } else if (arr[i] > secondGreatest && arr[i] < greatest) {
    secondGreatest = arr[i];
  }
}

console.log("Second greatest element:", secondGreatest);
```

Q100 Move Negatives Left and Positives Right

MEDIUM

```
let arr = [-1, 2, -3, 4, 5, -6, 7];
let left = 0;

for (let i = 0; i < arr.length; i++) {
  if (arr[i] < 0) {
    let temp = arr[i];
    arr[i] = arr[left];
    arr[left] = temp;
    left++;
  }
}

console.log("Rearranged array:", arr);
```

Q101 Left Rotation by K

MEDIUM

```
let arr = [1, 2, 3, 4, 5, 6, 7];
let k = 3;

k = k % arr.length; // handle k greater than array length

let rotated = arr.slice(k).concat(arr.slice(0, k));

console.log("After left rotation by " + k + ":", rotated);
```

Q102 Count Subarrays with Given Sum

MEDIUM

```
let arr = [1, 2, 3, 4, 5];
let targetSum = 5;
let count = 0;

for (let start = 0; start < arr.length; start++) {
  let sum = 0;
  for (let end = start; end < arr.length; end++) {
    sum += arr[end];
    if (sum === targetSum) count++;
  }
}

console.log("Number of subarrays with sum " + targetSum + ":", count);
```

Q103 Leaders in an Array

IMPORTANT

An element is a leader if it is greater than all elements to its right.

```
let arr = [16, 17, 4, 3, 5, 2];
let leaders = [];
let maxFromRight = arr[arr.length - 1];

leaders.push(maxFromRight); // last element is always a leader

for (let i = arr.length - 2; i >= 0; i--) {
  if (arr[i] > maxFromRight) {
    maxFromRight = arr[i];
    leaders.push(maxFromRight);
  }
}

leaders.reverse();
console.log("Leaders:", leaders);
```

Q104 Equilibrium Index

IMPORTANT

An index where the sum of elements on the left equals the sum on the right.

```
let arr = [-7, 1, 5, 2, -4, 3, 0];
let totalSum = arr.reduce((sum, val) => sum + val, 0);
let leftSum = 0;

for (let i = 0; i < arr.length; i++) {
  let rightSum = totalSum - leftSum - arr[i];
  if (leftSum === rightSum) {
    console.log("Equilibrium index:", i);
    break;
  }
  leftSum += arr[i];
}
```

Q105 Rearrange Positive and Negative Alternately

IMPORTANT

```
let arr = [1, 2, -3, -4, 5, -6];
let positives = arr.filter(num => num >= 0);
let negatives = arr.filter(num => num < 0);
let result = [];

let i = 0, j = 0;
while (i < positives.length && j < negatives.length) {
  result.push(positives[i++]);
  result.push(negatives[j++]);
}

// push any leftover elements
while (i < positives.length) result.push(positives[i++]);
while (j < negatives.length) result.push(negatives[j++]);

console.log("Rearranged array:", result);
```

Q106 Maximum Difference Between Two Elements

IMPORTANT

Find max(arr[j] - arr[i]) where j > i, so the larger element comes after the smaller one.

```
let arr = [2, 3, 10, 6, 4, 8, 1];
let minSoFar = arr[0];
let maxDiff = 0;

for (let i = 1; i < arr.length; i++) {
  if (arr[i] - minSoFar > maxDiff) {
    maxDiff = arr[i] - minSoFar;
  }
  if (arr[i] < minSoFar) {
    minSoFar = arr[i];
  }
}

console.log("Maximum difference:", maxDiff);
```

Q107 Union of Two Arrays

IMPORTANT

```
let arr1 = [1, 2, 3, 4];
let arr2 = [3, 4, 5, 6];

let unionSet = new Set([...arr1, ...arr2]);
let union = Array.from(unionSet);

console.log("Union:", union);
```

Q108 Intersection of Two Arrays

IMPORTANT

```
let arr1 = [1, 2, 3, 4];
let arr2 = [3, 4, 5, 6];

let set2 = new Set(arr2);
let intersection = arr1.filter(num => set2.has(num));

// remove duplicates from result
intersection = [...new Set(intersection)];

console.log("Intersection:", intersection);
```

Q109 Find Pair with Given Sum

IMPORTANT

```
let arr = [2, 7, 11, 15, 4];
let targetSum = 15;
let seen = new Set();
let pairFound = false;

for (let i = 0; i < arr.length; i++) {
  let needed = targetSum - arr[i];
  if (seen.has(needed)) {
    console.log("Pair found:", needed, "and", arr[i]);
    pairFound = true;
    break;
  }
  seen.add(arr[i]);
}

if (!pairFound) console.log("No pair found");
```

Q110 Maximum Product Subarray (Basic Version)

IMPORTANT

```
let arr = [2, 3, -2, 4];
let maxProduct = arr[0];

for (let start = 0; start < arr.length; start++) {
  let product = 1;
  for (let end = start; end < arr.length; end++) {
    product *= arr[end];
    if (product > maxProduct) {
      maxProduct = product;
    }
  }
}

console.log("Maximum product subarray:", maxProduct);
```